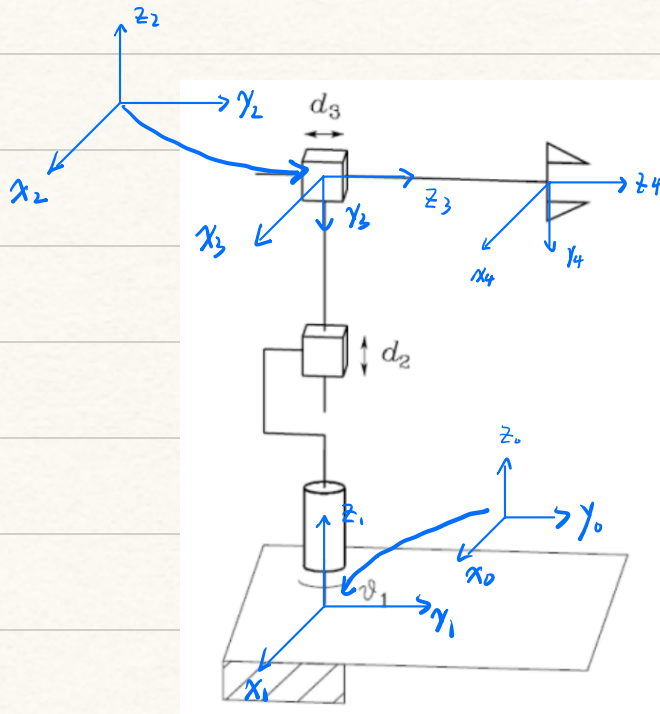


1.

1) D-H参数:

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	0	d_2	0
3	-90°	0	d_3	0
4	0	0	d_4	0



<建系方式不唯一>

2) 齐次变换矩阵:

$${}^0T_1 = \begin{pmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1T_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2T_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^3T_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

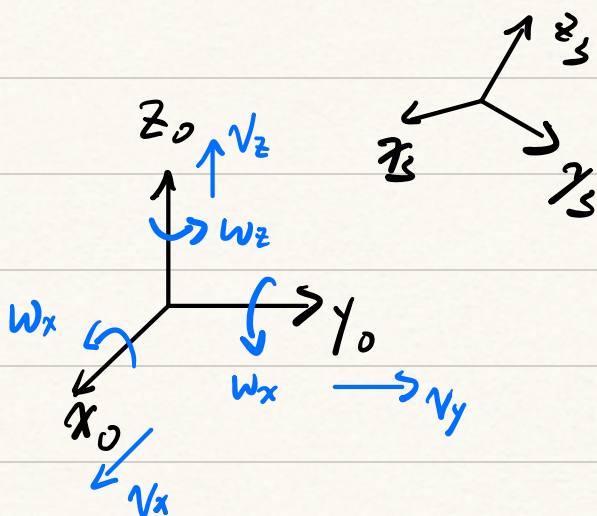
$${}^0T_3 = \begin{pmatrix} c_1 & 0 & -s_1 & -s_1 d_3 \\ s_1 & 0 & c_1 & c_1 d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0T_4 = \begin{pmatrix} c_1 & 0 & -s_1 & -s_1 d_4 - s_1 d_3 \\ s_1 & 0 & c_1 & c_1 d_4 + c_1 d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} s_1 & 0 & c_1 & c_1 d_3 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} s_1 & 0 & c_1 & c_1 d_4 + c_1 d_3 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

13)



$$\dot{\mathbf{x}} = \begin{pmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$

关节速度 $\dot{\theta} = (\dot{\theta}_1, \dot{d}_1, \dot{d}_2)$

<平动> 由 0T_3 得 $x_3(\theta) = -s_1 d_3$

$$\downarrow$$

$$\frac{\partial x_3}{\partial \theta} \cdot \dot{\theta} = \dot{x}_3 = v_x$$

$$\downarrow$$

$$\frac{\partial x_3}{\partial \theta} \cdot \dot{\theta} = \left(\frac{\partial x_3}{\partial \theta_1}, \frac{\partial x_3}{\partial d_1}, \frac{\partial x_3}{\partial d_2} \right) \cdot \begin{pmatrix} \dot{\theta}_1 \\ \dot{d}_1 \\ \dot{d}_2 \end{pmatrix} = v_x$$

Jacobian 第一行

<转动>

平面转动关节与移动关节

$$\begin{aligned} \omega &= \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} = \xi_1^0 z_1 \cdot \dot{\theta}_1 + \xi_2^0 z_2 \cdot \dot{\theta}_2 + \dots \\ &= \underbrace{(\xi_1^0 z_1, \xi_2^0 z_2, \dots,)}_{J^{3 \times N}} \cdot \begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \vdots \end{pmatrix} \end{aligned}$$

在本题中, $\theta = (\theta_1, d_2, d_3)$

$$\therefore \xi_2 = \xi_3 = 0, \quad \xi_1 = 1$$

$${}^0 z_1 = (0, 0, 1)^T$$

综上.

$$J = \begin{pmatrix} -c_1(d_3+d_4) & 0 & -s_1 \\ -s_1(d_3+d_4) & 0 & c_1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

2.

(1) D-H 参数

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	a_1	0	0	θ_2
3	a_2	0	d_3	0
4	0	0	0	θ_4
5	0	0	d_5	0

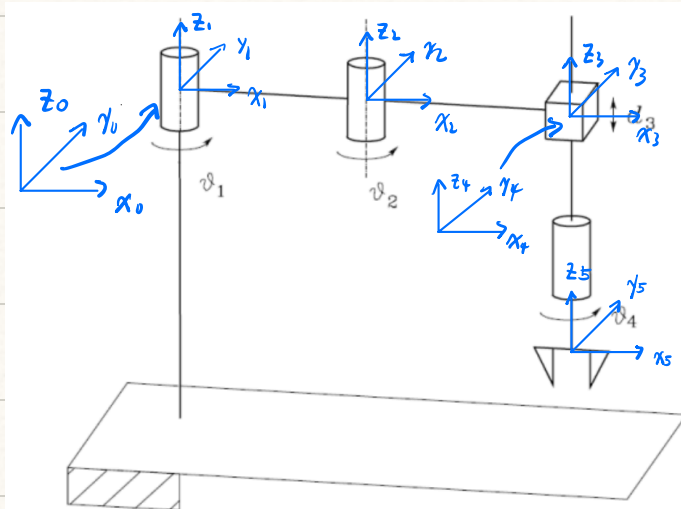


图 2

< 坐标系不唯一 >

(2) 齐次变换矩阵.

$${}^0T_1 = \begin{pmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1T_2 = \begin{pmatrix} c_2 & -s_2 & 0 & a_1 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0T_2 = \begin{pmatrix} c_1 c_2 - s_1 s_2 & -c_1 s_2 - s_1 c_2 & 0 & c_1 a_1 \\ s_1 c_2 + c_1 s_2 & -s_1 s_2 + c_1 c_2 & 0 & s_1 a_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2T_3 = \begin{pmatrix} 1 & 0 & 0 & a_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0T_3 = \begin{pmatrix} C_1C_2 - S_1S_2 & -C_1S_2 - S_1C_2 & 0 & a_2(C_1C_2 - S_1S_2) + C_1a_1 \\ S_1C_2 + C_1S_2 & -S_1S_2 + C_1C_2 & 0 & a_2(S_1C_2 + C_1S_2) + S_1a_1 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^3T_4 = \begin{pmatrix} C_4 & -S_4 & 0 & 0 \\ S_4 & C_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^4T_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0T_5 = (t_1, t_2, t_3, t_4)$$

$$t_1 = \begin{pmatrix} C_4(C_1C_2 - S_1S_2) - S_4(C_1S_2 + S_1C_2) \\ C_4(S_1C_2 + C_1S_2) - S_4(S_1S_2 - C_1C_2) \\ 0 \\ 0 \end{pmatrix}$$

$$t_2 = \begin{pmatrix} -S_4(C_1C_2 - S_1S_2) - C_4(C_1S_2 + S_1C_2) \end{pmatrix}$$

$$\begin{pmatrix} -s_4 (s_1 c_2 + c_1 s_2) - c_4 (s_1 s_2 - c_1 c_2) \\ 0 \\ 0 \end{pmatrix}$$

$$t_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$t_4 = \begin{pmatrix} a_2 (c_1 c_2 - s_1 s_2) + c_1 a_1 \\ a_2 (s_1 c_2 + c_1 s_2) + s_1 a_1 \\ d_3 + d_5 \\ 1 \end{pmatrix}$$

$$12) \quad J = \left(\frac{\partial x}{\partial \theta}, \frac{\partial y}{\partial \theta}, \frac{\partial z}{\partial \theta}, J_w \right)^T \quad \vec{\theta} = (\theta_1, \theta_2, d_3, \theta_4)^T$$

$$\frac{\partial x}{\partial \theta} = (a_2 (-s_1 c_2 - c_1 s_2) - s_1 a_1, a_2 (-c_1 s_2 - s_1 c_2), 0, 0)$$

$$\frac{\partial y}{\partial \theta} = (a_2 (c_1 c_2 - s_1 s_2) + c_1 a_1, a_2 (-s_1 s_2 + c_1 c_2), 0, 0)$$

$$\frac{\partial z}{\partial \theta} = (0, 0, 1, 0)$$

$$\xi_1 = \xi_2 = \xi_4 = 1 \quad \xi_3 = 0$$

$${}^0z_1 = {}^0z_2 = {}^0z_4 = (0, 0, 1)^T$$

$$J_w = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$